

Agentic Regulatory Intelligence & Compliance

Presented by BankSavers



PROBLEM STATEMENT

Modern banking institutions handle dozens of complex multi-jurisdictional updates daily. Manual tracking introduces catastrophic operational delays, critical blindspots, and massive regulatory penalties. This leaves compliance officers bogged down in paperwork, unable to extract clean actionable instructions from vast volumes of legal text.



Information Overload

Regulatory "noise" has ballooned, with constant overlapping feeds. Compliance teams drown in irrelevant data, making it difficult to pinpoint critical high-impact changes.



Translation gap

Opaque legal jargon cannot be directly executed by IT, HR, or Treasury. Converting complex mandates into deterministic, engineering-ready instructions remains slow and manual.



Accountability Vacuum

Once policies change, verifying compliance status manually is highly error-prone. The lack of structured evidence validation creates massive gaps before regulatory audits.

OUR SOLUTION

Sentinel-RegAI automates the transition from raw legal text to auditable workflows. Using dual-agent intelligence, it proactively detects anomalies and triggers deterministic validation checks. It cuts operating costs, ensures zero accountability gaps, and converts heavy regulatory texts into lightweight **Measurable Action Points (MAPs)**

Sentinel-RegAI operates on a dual specialized intelligence agent setup to secure the compliance cycle:

- **Agent Alpha: The Traffic Scanner**

A localized anomaly detection engine processing real-time RSS, PDF, and API feeds. It flags anomalous deviations and scores compliance risk.

- **Agent Beta: The Orchestrator**

An LLM-driven reasoning agent that breaks down alerts, schedules cross-departmental tickets, and performs dynamic validation checks.



SYSTEM ARCHITECTURE

Page 03



AGENT ALPHA: Traffic Scanner

Agent Alpha continuously monitors unstructured feeds to filter background noise from critical regulatory events.

- **Continuous Ingestion:** Feeds from the RBI, SEC, GDPR, and global regulatory boards.
- **NLP Transformer Classifier:** Evaluates deviations from Business As Usual (BAU) language.
- **Smart Escalation Trigger:** If the calculated Rscore exceeds the threshold θ , data immediately escalates to Agent Beta.

AGENT BETA: Orchestrator

Agent Beta turns complex, hard-to-interpret policies into explicit operational tasks using standard RAG architectures.

- **Generates Measurable Action Points (MAPs):** Conversational directives translate into concrete tasks (e.g., "Implement AES-256 validation criteria").
- **Skill-Matrix Association:** Maps tasks to departments (IT, HR, Legal, Treasury) using cosine similarity logic.
- **Continuous Output Auditing:** Periodically loops to audit document evidence submitted by employees.

Workflow Understanding



Requirements	Technical Description & Workflow
Multi-source Ingestion	Ingests data securely from global sources (RBI, SEC, GDPR, internal files) via raw web-scraping pipelines and robust secure APIs.
MAP Generation	Converts static declarations ("Improve privacy standards") into strict actionable instructions ("Deploy AES-256 for PII data fields").
Autonomous Assignment	Intelligently pairs specific compliance tasks with corresponding organizational departments based on an active skills matrix vector.
Evidence Validation	Collects validation documentation and tests evidence validity autonomously using NLP semantic similarity checks.



Multi-Phase Integration Roadmap

Page 05



01

Data Pipeline & Alpha Training

Build out raw multi-source ingestion pipelines and configure localized NLP anomaly model.

02

Beta Prompt Engineering

Establish LLM Chain-of-Thought prompts to convert raw rules to structured compliance MAPs.

03

Collaborative Integrations

Deploy automatic routing to JIRA, Slack, and internal ticketing hubs based on department skills.

04

Validation & Auditing

Launch the Semantic Analysis Validation Engine to autonomously verify finished tickets.

TECH STACK



Core ML Engine: (PyTorch / Hugging Face Transformers)

Using **Python 3.11+** combined with **PyTorch** and **Hugging Face Transformers** powers the anomaly scoring and transformer classification. No legacy scripts are allowed.



Vector Context: Pinecone or Milvus

Enterprise-grade vector database instances on **Pinecone** or **Milvus** store high-dimensional text embeddings, accelerating retrieval augmented compliance workflows.



Agentic Workflows: LangGraph or CrewAI

Complex tasks and state queues managed securely through **LangGraph** / **CrewAI**, utilizing robust **Celery with Redis** for distributed task queues.





THANK YOU

The Future of Banking Compliance

By shifting regulatory tracking from a reactive manual process to a proactive agentic loop, **Sentinel-RegAI** reduces operating overhead, guarantees compliance reliability, and empowers banks to thrive under complex multi-jurisdictional updates.

